Solution to Challenge 8

Two Incommensurate Frequencies Form a Torus

(Alligood, Sauer & Yorke, *Chaos: An Introduction to Dynamical Systems*, Ch. 8)

Overview. The Poincaré–Bendixson Theorem restricts limit sets of *autonomous planar* systems. This challenge shows what changes when we leave that setting: forcing an oscillator at a second frequency makes the planar picture nonautonomous, the trajectory can cross itself in the plane, and in the correct three-dimensional phase space the trajectory winds around a torus. When the two frequencies are incommensurate, the trajectory never closes and fills the torus densely.

Throughout, a dot denotes d/dt. I use freely the Existence and Uniqueness Theorem for ODEs (linear equations with constant coefficients and smooth forcing certainly satisfy it).

Step 1: Solutions of $\ddot{x} + 4x = 0$

Claim. Every solution has the form $x(t) = a\sin(2t + b)$, where a, b are determined by $x(0) = x_0$, $\dot{x}(0) = v_0$.

Proof. Every function of this form is a solution: $\dot{x} = 2a\cos(2t+b)$ and $\ddot{x} = -4a\sin(2t+b) = -4x$.

We can match any initial condition. Expanding,

$$x(0) = a\sin b, \qquad \dot{x}(0) = 2a\cos b.$$

Setting $a\sin b = x_0$ and $a\cos b = v_0/2$ we get

$$a = \sqrt{x_0^2 + \tfrac{1}{4}v_0^2} \;\geq 0, \qquad \sin b = \frac{x_0}{a}, \quad \cos b = \frac{v_0}{2a} \quad (\text{if } a > 0),$$

and these two equations determine b modulo 2π (since $(\sin b, \cos b)$ is a point on the unit circle). If $a = 0$ then $x_0 = v_0 = 0$, the solution is $x \equiv 0$, and b is arbitrary.

No other solutions exist. For the initial value problem, the Existence and Uniqueness Theorem says there is exactly one solution, so the function $a\sin(2t + b)$ constructed above *is* the solution. (Equivalently: the characteristic equation $r^2 + 4 = 0$ gives $r = \pm 2i$, so the general solution is $A\cos 2t + B\sin 2t$, and $A\cos 2t + B\sin 2t = a\sin(2t+b)$ with $a = \sqrt{A^2 + B^2}$, $a\sin b = A$, $a\cos b = B$.) ■

Step 2: The first-order system and its phase portrait

Put $y = \dot{x}$. Then

$$\dot{x} = y, \qquad \dot{y} = -4x, \tag{1}$$

an *autonomous* system in $\mathbb{R}^2$, so the Poincaré–Bendixson Theorem applies. By Step 1, with $x = a\sin(2t + b)$,

$$(x(t), y(t)) = \big(a\sin(2t + b),\ 2a\cos(2t + b)\big).$$

If $a > 0$, then

$$\frac{x^2}{a^2} + \frac{y^2}{(2a)^2} = \sin^2(2t+b) + \cos^2(2t+b) = 1,$$

so the orbit lies on the ellipse with semi-axes a (in x) and $2a$ (in y). (Alternatively, $E = 4x^2 + y^2$ is conserved: $\dot{E} = 8x\dot{x} + 2y\dot{y} = 8xy - 8xy = 0$.) As t runs over one period, $2t + b$ runs over all angles in $[0, 2\pi)$, so the orbit covers the whole ellipse, traversed clockwise. The point returns to its starting position when $2t$ increases by 2π, i.e. after time π. So the orbit is a periodic orbit of period π, i.e. of angular frequency

$$\frac{2\pi}{\text{period}} = \frac{2\pi}{\pi} = 2 \text{ radians per unit time.}$$

The origin ($a = 0$) is the only equilibrium; all other orbits are nested ellipses, each its own ω-limit set, in agreement with Poincaré–Bendixson.

Step 3: The forced equation

Solve

$$\ddot{x} + 4x = 3\sin t, \qquad x(0) = 0,\ \dot{x}(0) = 3. \tag{2}$$

The forcing frequency 1 differs from the natural frequency 2, so there is no resonance and we can look for a particular solution $x_p = C\sin t$. Then $\ddot{x}_p + 4x_p = (-C + 4C)\sin t = 3C\sin t$, so $C = 1$ and $x_p = \sin t$. The general solution is

$$x(t) = \sin t + A\cos 2t + B\sin 2t.$$

Initial conditions:

$$x(0) = A = 0, \qquad \dot{x}(0) = \cos 0 + 2B\cos 0 = 1 + 2B = 3 \ \Rightarrow\ B = 1.$$

Therefore

$$\boxed{x(t) = \sin t + \sin 2t}. \tag{3}$$

Check: $\ddot{x} = -\sin t - 4\sin 2t$, so $\ddot{x} + 4x = 3\sin t$. ✓

Step 4: The trajectory in the plane crosses itself

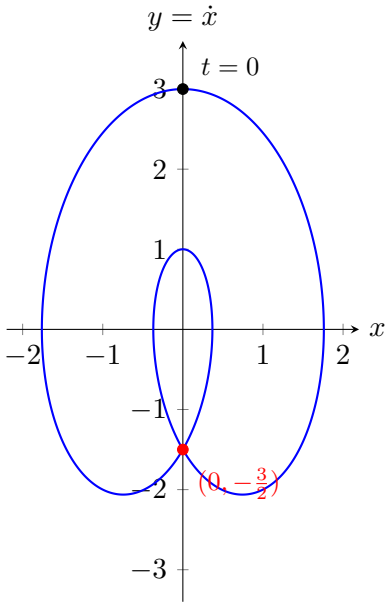
As a first-order system, (2) reads

$$\dot{x} = y, \qquad \dot{y} = -4x + 3\sin t,$$

whose right side depends explicitly on t: it is *nonautonomous*, so Poincaré–Bendixson does not apply. The solution curve is

$$(x(t), y(t)) = \big(\sin t + \sin 2t,\ \cos t + 2\cos 2t\big), \qquad 0 \le t \le 2\pi,$$

and it is periodic with period 2π.



Exhibiting the self-crossing. Note $x(-t) = -x(t)$ and $y(-t) = y(t)$, so the curve is symmetric about the y-axis. Solve $x(t) = 0$:

$$\sin t + \sin 2t = \sin t\,(1 + 2\cos t) = 0 \implies t \in \{0, \pi\} \text{ or } \cos t = -\tfrac{1}{2}, \text{ i.e. } t = \tfrac{2\pi}{3}, \tfrac{4\pi}{3}.$$

At these last two times,

$$y\big(\tfrac{2\pi}{3}\big) = \cos\tfrac{2\pi}{3} + 2\cos\tfrac{4\pi}{3} = -\tfrac{1}{2} - 1 = -\tfrac{3}{2}, \qquad y\big(\tfrac{4\pi}{3}\big) = \cos\tfrac{4\pi}{3} + 2\cos\tfrac{8\pi}{3} = -\tfrac{1}{2} - 1 = -\tfrac{3}{2}.$$

So the curve passes through the point $(0, -\frac{3}{2})$ at two different times in $[0, 2\pi)$, $t_1 = 2\pi/3$ and $t_2 = 4\pi/3$. The crossing is transversal: the velocity vectors $(\dot{x}, \dot{y}) = (y, \ddot{x})$ there are

$$\ddot{x}(t) = -\sin t - 4\sin 2t: \qquad \ddot{x}(t_1) = \tfrac{3\sqrt{3}}{2}, \qquad \ddot{x}(t_2) = -\tfrac{3\sqrt{3}}{2},$$

so the tangents are $(-\frac{3}{2},\ \pm\frac{3\sqrt{3}}{2})$, which are not parallel.

Is this consistent with what we have learned? Yes. For an autonomous planar system, uniqueness of solutions means that two orbits through the same point coincide, so a periodic orbit is a simple closed curve and orbits cannot cross. For (2) the velocity at a point (x, y) depends on *when* the trajectory is there, namely $\ddot{x} = -4x + 3\sin t$; at $(0, -\frac{3}{2})$ it is $3\sin t_1 = +\frac{3\sqrt{3}}{2}$ at t_1 and $3\sin t_2 = -\frac{3\sqrt{3}}{2}$ at t_2. No contradiction arises: uniqueness holds in the extended space (x, y, t), where there is no crossing; only the projection to the (x, y)-plane crosses itself.

Claim. $x(t) = \sin t + \sin 2t$ is not a solution of *any* equation

$$\ddot{x} + a\dot{x} + bx = f(x), \qquad x(0) = x_0,\ \dot{x}(0) = x_1 \tag{8.6}$$

(with a, b constants and f any function of x alone).

Proof. Suppose it were. Then at every time t,

$$\ddot{x}(t) = -a\,\dot{x}(t) - b\,x(t) + f(x(t)) = F\big(x(t), \dot{x}(t)\big),$$

where $F(x, y) = -ay - bx + f(x)$ is a fixed function of the *state* (x, y). Hence the acceleration is determined by the position $(x, \dot{x})$ in the plane. But we showed $(x(t_1), \dot{x}(t_1)) = (x(t_2), \dot{x}(t_2)) = (0, -\frac{3}{2})$ while $\ddot{x}(t_1) = \frac{3\sqrt{3}}{2} \neq -\frac{3\sqrt{3}}{2} = \ddot{x}(t_2)$. Both would equal $F(0, -\frac{3}{2}) = \frac{3}{2}a + f(0)$, a contradiction. ■

Remark. The same argument rules out every autonomous equation $\ddot{x} = F(x, \dot{x})$. Equivalently: if $x(t)$ solved an autonomous equation, then by uniqueness the shifted function $x(t + t_2 - t_1)$ would solve the same initial value problem as $x(t)$ at time t_1, forcing x to have period $t_2 - t_1 = 2\pi/3$, whereas the minimal period of $\sin t + \sin 2t$ is 2π.

Step 5: The general family $x(t) = \sin t + \sin ct$

Let $c \in \mathbb{R}$. Then $\ddot{x} = -\sin t - c^2 \sin ct$, so

$$\ddot{x} + c^2 x = -\sin t - c^2 \sin ct + c^2 \sin t + c^2 \sin ct = (c^2 - 1) \sin t,$$

which is (2) generalized (for $c = 2$ it gives $3 \sin t$). Initial conditions: $x(0) = \sin 0 + \sin 0 = 0$ and $\dot{x}(0) = \cos 0 + c \cos 0 = c + 1$. ✓ Thus $x(t) = \sin t + \sin ct$ solves

$$\ddot{x} + c^2 x = (c^2 - 1) \sin t, \quad x(0) = 0,\ \dot{x}(0) = c + 1,$$

and by uniqueness it is the solution. (The case $c = 0$ gives $x = \sin t$, $\ddot{x} = -\sin t$, and $c = 1$ gives $x = 2 \sin t$, both consistent.)

Step 6: An autonomous system in $\mathbb{R}^3$

Consider

$$\dot{x} = y, \qquad \dot{y} = -c^2 x + (c^2 - 1) \sin \theta, \qquad \dot{\theta} = 1, \tag{4}$$

with $x(0) = 0,\ y(0) = c + 1,\ \theta(0) = 0$. Time has become a coordinate, θ, which makes the system autonomous.

Claim.

$$x(t) = \sin t + \sin ct, \qquad y(t) = \cos t + c \cos ct, \qquad \theta(t) = t$$

is the solution.

Verification.

- $\dot{x} = \cos t + c \cos ct = y$. ✓
- $\dot{\theta} = 1$ and $\theta(0) = 0$ give $\theta = t$. ✓
- $\dot{y} = -\sin t - c^2 \sin ct$, while

$$-c^2 x + (c^2 - 1) \sin \theta = -c^2 \sin t - c^2 \sin ct + c^2 \sin t - \sin t = -\sin t - c^2 \sin ct. ✓$$

- Initial values: $x(0) = 0$, $y(0) = 1 + c$, $\theta(0) = 0$. ✓

The right-hand side of (4) is smooth, so by uniqueness this is *the* solution. ■

Step 7: Commensurate frequencies give a periodic orbit

Since θ only matters modulo 2π, the actual phase space is $\mathbb{R}^2 \times (\mathbb{R}/2\pi\mathbb{Z})$, and the trajectory is

$$\big(\sin t + \sin ct,\ \cos t + c \cos ct,\ t \bmod 2\pi\big).$$

Suppose 1 and c are commensurate: $c = p/q$ with integers $p \neq 0$, $q \geq 1$, $\gcd(p, q) = 1$ (the case $c = 0 = 0/1$ is also covered; negative p is allowed).

Claim. The trajectory is a periodic orbit of (least) period $T = 2\pi q$.

Proof. $2\pi q$ *is a period.* For all t,

$$\sin(t+2\pi q)=\sin t, \qquad \sin\big(c(t+2\pi q)\big)=\sin(ct+2\pi p)=\sin ct,$$
$$\cos(t+2\pi q)=\cos t, \qquad \cos\big(c(t+2\pi q)\big)=\cos(ct+2\pi p)=\cos ct,$$

and $\theta = t+2\pi q \equiv t \pmod{2\pi}$. So the point at time $t+2\pi q$ equals the point at time t.

No smaller positive time works. Suppose the trajectory is back at its initial point $(0,\,1+c,\,0)$ at time $T>0$. The θ-coordinate forces $T=2\pi m$ for some positive integer m. The y-coordinate then gives

$$y(T)=\cos 2\pi m + c\cos\big(2\pi mp/q\big)=1+c\cos\big(2\pi mp/q\big).$$

Requiring $y(T)=1+c$ and $c\neq 0$ gives $\cos(2\pi mp/q)=1$, i.e. $mp/q\in\mathbb{Z}$. Since $\gcd(p,q)=1$, this forces $q\mid m$, so $m\geq q$ and $T\geq 2\pi q$. (For $c=0$, $q=1$ and the condition $T=2\pi m$ already gives least period $2\pi=2\pi q$.)

Because (4) is autonomous, returning to the initial point at time T makes the solution periodic with period T. Hence the least period is $2\pi q$. ■

Remark. During one period the θ-angle goes around q times and the second oscillation $\sin ct$ completes p cycles ($|p|$ if $c<0$): the orbit is a closed curve winding q times in one direction and p times in the other on the torus described next. Note that $2\pi q$ is the common period of $\sin t$ (period 2π) and $\sin ct$ (period $2\pi q/|p|$).

Step 8: Irrational c: the torus

Now let c be irrational (in particular $c\neq 0$). Fix $\theta_0\in[0,2\pi)$. The trajectory is in the plane $\theta=\theta_0$ (modulo 2π) exactly at the times $t_k=\theta_0+2\pi k$, $k\in\mathbb{Z}$. At these times

$$x(t_k)=\sin(\theta_0+2\pi k)+\sin(ct_k)=\sin\theta_0+\sin\varphi_k,$$
$$y(t_k)=\cos(\theta_0+2\pi k)+c\cos(ct_k)=\cos\theta_0+c\cos\varphi_k,$$

where

$$\varphi_k := ct_k = c\theta_0+2\pi ck.$$

Therefore

$$\big(x(t_k)-\sin\theta_0\big)^2+\frac{\big(y(t_k)-\cos\theta_0\big)^2}{c^2}=\sin^2\varphi_k+\cos^2\varphi_k=1,$$

so the points lie on the ellipse

$$E_{\theta_0}: \quad (x-\sin\theta_0)^2+\frac{(y-\cos\theta_0)^2}{c^2}=1,$$

centered at $(\sin\theta_0,\cos\theta_0)$ with semi-axes 1 (horizontal) and $|c|$ (vertical). ■

The ellipses fit together into a torus. Define

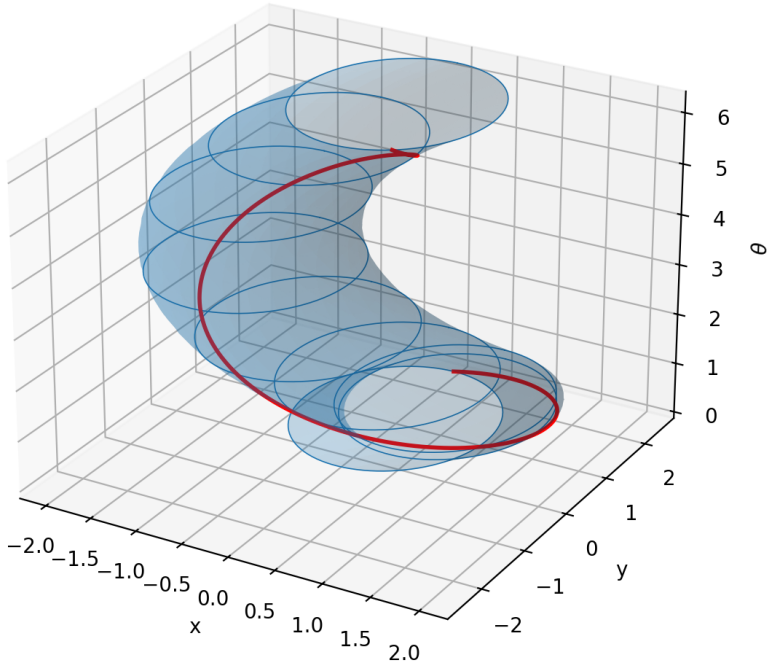
$$\Phi(\theta,\varphi)=\big(\sin\theta+\sin\varphi,\ \cos\theta+c\cos\varphi,\ \theta\big), \qquad (\theta,\varphi)\in(\mathbb{R}/2\pi\mathbb{Z})^2.$$

For each fixed θ this traces out the ellipse E_θ as φ goes around once, and the trajectory is simply $\Phi(t,ct)$. Let $\mathcal{T}=\Phi\big((\mathbb{R}/2\pi\mathbb{Z})^2\big)=\bigcup_\theta E_\theta\times\{\theta\}$.

- Φ *is one-to-one.* The θ-coordinate recovers θ, and then $(\sin\varphi, c\cos\varphi)$ determines $(\sin\varphi,\cos\varphi)$ because $c\neq 0$, hence $\varphi \bmod 2\pi$.

- Φ *is a smooth immersion.* $\partial_\theta \Phi = (\cos\theta, -\sin\theta, 1)$ and $\partial_\varphi \Phi = (\cos\varphi, -c\sin\varphi, 0)$ are linearly independent ($\partial_\varphi \Phi \neq 0$ since $c \neq 0$, and only $\partial_\theta \Phi$ has a nonzero third component).
- A one-to-one continuous map from a compact space into a Hausdorff space is a homeomorphism onto its image.

Hence $\mathcal{T}$ is a two-dimensional torus (embedded smoothly in $\mathbb{R}^2 \times S^1$): a circle of ellipses E_θ, whose centers $(\sin\theta, \cos\theta)$ travel once around the unit circle as θ goes from 0 to 2π, glued by $\theta = 0 \sim 2\pi$. The vector field (4) is tangent to $\mathcal{T}$ (since the trajectory lies on it), so $\mathcal{T}$ is invariant.



Torus $\mathcal{T}$ for $c = \sqrt{2}$, drawn with $\theta \in [0, 2\pi]$ as the vertical axis (top and bottom ellipses are identified); the trajectory over one turn in θ ($0 \le t \le 2\pi$) is in red; it then continues from the bottom ellipse at a shifted point, never closing up.

The trajectory never closes and is dense on $\mathcal{T}$ (the Postscript's claim).

Not periodic. A return to the initial point at time $T > 0$ requires $T = 2\pi m$ ($m \in \mathbb{N}$) from θ, and then $\varphi = cT = 2\pi mc \equiv 0 \pmod{2\pi}$, i.e. $mc \in \mathbb{Z}$. This is impossible for $m \geq 1$ because c is irrational.

Dense. On the circle $\mathbb{R}/2\pi\mathbb{Z}$, the angles $\varphi_k = c\theta_0 + 2\pi ck$ are the orbit of the rotation by $2\pi c$, i.e. by the fraction c of a full turn. Since c is irrational, the φ_k are pairwise distinct (if $\varphi_j = \varphi_k$ then $c(j-k) \in \mathbb{Z}$). Given $\varepsilon > 0$, by the pigeonhole principle two of the first $N > 2\pi/\varepsilon$ points $\varphi_1, \ldots, \varphi_N$ lie within ε of each other. Their difference is a nonzero rotation by an angle δ with $|\delta| < \varepsilon \pmod{2\pi}$, and repeated application of it moves φ_k in steps of size less than ε all the way around the circle. Thus every arc of length ε contains some φ_k, so $\{\varphi_k\}$ is dense on the circle. Applying $\Phi(\theta_0, \cdot)$, the trajectory is dense on each ellipse E_{θ_0}, hence dense on the torus $\mathcal{T}$.

Interpretation. The map $\varphi \mapsto \varphi + 2\pi c$ on the cross-section circle E_{θ_0} is the Poincaré map: a rotation incommensurate with the circumference (the quasiperiodic circle map of Chapter 3). The trajectory is *quasiperiodic*, with two incommensurate frequencies 1 and c, and its ω-limit set is the entire torus. A two-dimensional invariant surface that is an ω-limit set cannot occur for planar autonomous systems, which shows that the Poincaré–Bendixson conclusions fail once we go to three dimensions.

Summary

Step	Result
1–2	$\ddot{x} + 4x = 0$: $x = a\sin(2t + b)$; orbits in $(x, \dot{x})$ are ellipses of period π (frequency 2).
3	$\ddot{x} + 4x = 3\sin t$, $x(0) = 0$, $\dot{x}(0) = 3$ has solution $x = \sin t + \sin 2t$.
4	Its planar trajectory self-crosses at $(0, -\frac{3}{2})$ ($t = \frac{2\pi}{3}, \frac{4\pi}{3}$); no equation $\ddot{x} + a\dot{x} + bx = f(x)$ has it as a solut
5–6	$x = \sin t + \sin ct$ solves (4), an autonomous system in $\mathbb{R}^3$.
7	$c = p/q$ in lowest terms: periodic orbit of period $2\pi q$.
8	c irrational: trajectory lies on, and is dense in, the torus $\bigcup_\theta E_\theta$; it never repeats.